THE INCOMPLETENESS OF OBSERVATION

Why Quantum Mechanics and General Relativity Cannot Be Unified From Within

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Abstract

An observer embedded in a deterministic universe cannot access the complete state. We prove that any such observer — coupled to a slow, high-capacity hidden sector on a finite-dimensional configuration space — must describe the visible sector using P-indivisible stochastic dynamics, mathematically equivalent to unitary quantum mechanics. The converse also holds: any quantum system, realized as a deterministic dilation, requires non-trivial coupling, slow-bath memory, and sufficient hidden-sector capacity. The Schrödinger equation, Born rule, and Bell violations emerge as structural consequences requiring no independent quantum postulates.

Applied to the cosmological horizon, the framework uniquely determines $\hbar = c^3\epsilon^2/(4G)$, fixes $\epsilon = 2\,l_p$, and recovers the Bekenstein-Hawking entropy $S = A/(4\,l_p^2)$. The 10^{122} cosmological constant discrepancy is identified as S_{dS}, the information compression ratio of the emergent quantum description; the observed vacuum energy is the mandatory classical baseline. Falsifiable predictions include dark energy evolution in Running Vacuum Model form with $\nu_{\mathrm{OI}} \approx 2.45 \times 10^{-3}$ (parametric; systematic range 1–5 $\times\, 10^{-3}$), consistent with current DESI data, and gravitational wave echoes near black hole horizons. The trace-out that produces QM simultaneously renders ~95% of the universe's gravitational budget invisible to the emergent description — matching the observed dark sector — providing independent corroboration of observational incompleteness.

1 PART I: THE GENERAL THEOREM

1.1 1. INTRODUCTION

1.1.1 1.1 The Problem of Embedded Observation

An observer embedded in a deterministic universe cannot access the complete state: degrees of freedom beyond causal reach are permanently hidden, and the observer's description is obtained by marginalizing over the hidden sector. The question is what this reduction imposes on the form of the observer's physical laws.

Prior work (QBism, relational QM, 't Hooft's cellular automaton [1]) takes observer-dependence as an interpretive starting point or derives it from specific microphysical models. This framework differs by identifying *necessary and sufficient conditions* under which any embedded observer in any deterministic system necessarily sees quantum mechanics — and proving these are the *only* conditions under which QM arises from a deterministic embedding. These constraints carry empirical weight: the framework reveals that phenomena previously viewed as catastrophic anomalies — most notably the 10^{122} cosmological constant discrepancy — are mandatory consequences of embedded observation.

1.1.2 1.2 Axioms

The framework rests on four axioms referencing no quantum mechanics, general relativity, or specific physical theory.

1. **Deterministic dynamics.** The total system evolves deterministically on a phase space Γ via a Hamiltonian flow ϕ_t. A probability density ρ on Γ (representing the observer's incomplete knowledge) evolves by the Liouville equation:

$$\frac{\partial \rho}{\partial t} = \{H_{\text{tot}}, \rho\} \equiv \mathcal{L}\rho$$

2. **Finiteness.** The effective configuration space of the visible sector is finite-dimensional, with a discreteness scale ϵ providing a finite minimal cell volume. Any observer bounded by a finite-area surface can couple to only finitely many modes across that boundary; independent support comes from holographic entropy bounds [27]. In Part I, ϵ is free; self-consistency (§6) pins $\epsilon = 2\,l_p$.

3. **Causal partition.** Γ is partitioned into a visible sector Γ_V and hidden sector Γ_H:

$$\Gamma = \Gamma_V \times \Gamma_H, \qquad H_{\text{tot}} = H_V + H_H + H_{\text{int}}$$

Cross-partition correlations enter only through H_{int}. The product decomposition is idealized; §4 and §10.7 address approximation quality and the conditions under which the sharp partition is exact.

4. **Classical probability.** The observer's predictions are classical expectation values over phase space; standard probability theory applies.

These axioms contain no quantum postulates. The claim is that quantum mechanics *emerges* under the conditions below.

1.1.3 1.3 Conditions on the Hidden Sector

(C1) Non-zero coupling. $H_{\text{int}} \neq 0$. The coupling is bidirectional.

(C2) Slow-bath timescale separation. The visible-sector (system) timescale τ_S is much shorter than the hidden-sector (bath) timescale τ_B: $\tau_S \ll \tau_B$ — the *inverse* of the Markovian regime. The hidden sector evolves on timescales far exceeding those accessible to the observer.

(C3) Sufficient capacity. The number of hidden-sector degrees of freedom N_H exceeds the number of visible-sector degrees of freedom N_V by enough that visible-sector interactions do not appreciably perturb the hidden sector's state on timescales $\ll \tau_B$.

Theorem statement. Under Axioms 1–4 and (C1)–(C3), the embedded observer's reduced description is mathematically equivalent to unitarily evolving quantum mechanics. The conditions are not merely sufficient but *necessary* (§3.3), establishing full equivalence.

1.1.4 1.4 Partition-Relativity

Lemma. *The emergent description is uniquely determined by the partition. Any parameters of the emergent theory depend only on the geometric and thermodynamic properties of the partition boundary.*

Proof. By Axiom 1, the Hamiltonian flow ϕ_t is unique. By Axiom 3, the partition is fixed. By Axiom 4, the observer uses the Liouville measure μ — the unique absolutely continuous invariant measure on the full phase space. (An embedded observer with incomplete knowledge of the total energy averages over energies with a smooth prior, recovering Liouville on a phase-space band; singular invariant measures are excluded by Axiom 4.) Letting π_V denote projection onto the visible sector, the marginalized transition probabilities

$$T_{ij}(t_2, t_1) = \int_{\Gamma_H} \delta_{x_j}[\pi_V(\phi_{t_2-t_1}(x_i, h))]\, d\mu(h)$$

are therefore uniquely determined by three inputs: dynamics (ϕ_t), partition (Γ_V, Γ_H), and measure (μ). Since ϕ_t and μ are fixed by the axioms, the stochastic process — and hence any emergent quantum description (§3.1) — depends only on the partition. □

1.2 2. EMERGENT STOCHASTICITY AND P-INDIVISIBILITY

1.2.1 2.1 Emergent Stochasticity

The total system evolves deterministically, but the visible sector alone is stochastic. The observer knows x but not h; different hidden states h_k compatible with the same x send it to different futures x'_k. Transition probabilities are obtained by marginalizing over the Liouville measure (§1.4). This makes the framework a hidden variable theory; consistency with Bell's theorem is addressed in §3.2.

1.2.2 2.2 The Slow-Bath Regime

The Markovian limit requires $\tau_B \ll \tau_S$ [4]. Condition (C2) inverts this: $\tau_S \ll \tau_B$. A slow bath must be distinguished from a static field. By (C1), coupling is continuously active: each visible-sector transition imprints on the hidden sector. Because the hidden sector is slow (not static), imprints persist without thermal overwriting. On subsequent transitions, coupling reads back stored correlations, producing history-dependent transition probabilities — the non-Markovian regime [4].

1.2.3 2.3 P-Indivisibility

By Axiom 2, the visible and hidden sectors have finite configuration spaces $\mathcal{C}_V$ and $\mathcal{C}_H$ (the discrete counterparts of Γ_V and Γ_H). On these finite sets:

Theorem. *Let $\mathcal{C}_V$ and $\mathcal{C}_H$ be finite sets with $|\mathcal{C}_V| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection. Define:*

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi(x_i, h)) = x_j\}|}{|\mathcal{C}_H|}$$

and the k-step matrix $T_{ij}^{(k)}$ by applying φ^k. If T is not a permutation matrix, then the process is P-indivisible.

Proof. Uses total variation distance $d(p, q) = \frac{1}{2}\sum_k |p_k - q_k|$. P-divisibility $\iff$ $d(p(t), q(t))$ non-increasing for all initial p, q [6, 7].

Step 1 (Recurrence). φ bijective on a finite set $\Rightarrow \exists N : \varphi^N = \mathrm{id}$. Thus $T^{(N)} = I$ and $d(\delta_i T^{(N)}, \delta_j T^{(N)}) = 1$ for $i \neq j$.

Step 2 (Strict contraction). T not a permutation $\Rightarrow \exists i, j, l$: $T_{il} > 0$ and $T_{jl} > 0$. Then:

$$d(\delta_i T, \delta_j T) = \frac{1}{2}\sum_k |T_{ik} - T_{jk}| < 1$$

Step 3. $d(\delta_i T^{(1)}, \delta_j T^{(1)}) < 1 = d(\delta_i T^{(N)}, \delta_j T^{(N)})$: non-monotonic, hence P-indivisible. □

The theorem requires only bijective dynamics (Axiom 1) and non-trivial coupling (C1). Axiom 2 guarantees the finite configuration space.

Convergence with prior work. The theorem converges with independent results: Pechukas [5] (reduced dynamics need not preserve positivity), Rivas et al. [6, 7] (divisibility failure $\leftrightarrow$ information backflow), Pollock et al. [8] (process-tensor Markovianity conditions), and Strasberg and Esposito [9] (non-Markovian reduced dynamics in the slow-bath regime of C2). Bylicka, Johansson, and Acín [34] prove that for bijective dynamical maps — the class generated here — CP-divisibility and information backflow are equivalent, so the P-indivisibility established above implies CP-indivisibility directly. Milz et al. [35] further show that even CP-divisibility does not guarantee Markovianity in the process-tensor sense, confirming that the non-Markovian character identified here is robust under all current definitions.

Continuous-time extension. The Hamiltonian flow on finite-dimensional phase space preserves Liouville measure on compact energy surfaces. $T_{ij}(t)$ is continuous with $T(0) = I$. By (C1), $T(t)$ departs from the permutation class for $t > 0$. By Poincaré recurrence, $\exists t_R$: the set of hidden states with $\pi_V(\varphi_{t_R}(x_i, h)) = x_i$ has measure $> 1 - \delta$ for any $\delta > 0$. For small δ, this gives non-monotonic trace distance, establishing P-indivisibility in continuous time.

The recurrence argument establishes P-indivisibility in principle. The following lemma shows that under (C2), information backflow occurs on observable timescales — not merely at Poincaré recurrence times.

Lemma (Accessible-timescale backflow). *Under (C1)–(C3) with $\tau_S \ll \tau_B$, the non-Markovian mutual information $I(X_{<t}; X_{>t} \mid X_t)$ is $\mathcal{O}(1)$ for observation windows $t \sim k\tau_S$ with $k\tau_S \ll \tau_B$.*

Proof. The coupling H_{int} transfers visible-sector information to the hidden sector at each interaction, at rate $\sim 1/\tau_S$. Between interactions, the hidden sector evolves under H_H with spectral

gap $\Delta \sim 1/\tau_B$. The decay of correlations stored in the hidden sector is governed by $e^{-\Delta\tau_S}$. When $\tau_S \ll \tau_B$, $\Delta\tau_S \ll 1$, so the decay per visible-sector transition is $1 - e^{-\Delta\tau_S} \approx \Delta\tau_S \ll 1$: correlations survive each step essentially intact. After k transitions spanning a time $k\tau_S \ll \tau_B$, the cumulative decay is $e^{-k\Delta\tau_S} \approx 1 - k\Delta\tau_S$, which remains close to unity. The hidden sector therefore retains $\sim k$ bits of visible-sector history over this window, and the $(k+1)$-th transition reads back stored correlations through H_{int}, producing history-dependent transition probabilities — i.e., information backflow. More precisely, the mutual information satisfies:

$$I(X_{<t}; X_{>t} \mid X_t) \geq I_0(1 - k\Delta\tau_S) = I_0\left(1 - \frac{k\tau_S}{\tau_B}\right)$$

where $I_0 > 0$ is the single-step information transfer from (C1). For $k\tau_S \ll \tau_B$, this remains $\mathcal{O}(I_0)$ — comparable to the single-step coupling strength, not exponentially suppressed. The bound saturates (C3): the maximum storable history is $\log_2 m$ bits (§3.3), so persistent backflow over K transitions requires $m \geq 2^K$, which is amply satisfied when $N_H \gg N_V$. □

Role of (C2) and (C3). The Poincaré recurrence argument and the accessible-timescale lemma are independent: the former establishes P-indivisibility from (C1) and finiteness alone; the latter shows that (C2) and (C3) promote it from a formal property to an observationally dominant one.

Weak-coupling and fast-bath regimes. Known P-divisible systems with non-zero coupling illustrate the necessity of (C2), not a failure of the theorem. The Jaynes-Cummings model with Lorentzian spectral density is P-divisible when dissipation dominates coupling — the fast-bath regime ($\tau_B \ll \tau_S$), violating (C2). The Davies-Merkli Markovian convergence theorem likewise applies in the Born-Markov (fast-bath) limit. The §3.3 necessity proof for (C2) establishes this formally.

1.2.4 2.4 Explicit Construction: A Minimal Model

Visible sector: $x \in \{0, 1\}$. Hidden sector: $h \in \{1, \ldots, 6\}$. Dynamics: the permutation

$$\sigma = (0{,}1 \leftrightarrow 1{,}1)\ (0{,}2 \leftrightarrow 1{,}2)\ (0{,}3 \leftrightarrow 0{,}4)\ (0{,}5 \leftrightarrow 0{,}6)\ (1{,}3 \leftrightarrow 1{,}4)\ (1{,}5 \leftrightarrow 1{,}6)$$

satisfying (C1)–(C3). Since $\sigma^2 = \text{id}$: at $t = 1$, two of six hidden states flip x, giving

$$T(1, 0) = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}$$

At $t = 2$ ($\sigma^2 = \text{id}$): $T(2, 0) = I$. A Markov chain would predict continued mixing: $T(1, 0)^2 = \left(\begin{smallmatrix} 5/9 & 4/9 \\ 4/9 & 5/9 \end{smallmatrix}\right)$. The actual dynamics show complete *un-mixing*. Computing the intermediate propagator:

$$\Lambda(2, 1) = T(2, 0) \cdot [T(1, 0)]^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Negative entries — no valid stochastic matrix exists. **P-indivisible.** □

The mechanism. The hidden sector acts as a memory register. The transpositions $(0, h) \leftrightarrow (1, h)$ for $h \in \{1, 2\}$ flip x while preserving h, so a state at $x = 1$ with $h \in \{1, 2\}$ after step 1 carries the record "I was at $x = 0$." Step 2 reads this record and flips x back — information

backflow impossible for a memoryless process. The cosmological realization has $\tau_S/\tau_B \sim 10^{-32}$; P-indivisibility is more extreme, not less, when memory is more persistent.

1.3 3. THE EMERGENCE OF QUANTUM MECHANICS

1.3.1 3.1 The Stochastic-Quantum Correspondence

By Barandes' correspondence [10, 11], any P-indivisible stochastic process on a finite configuration space of size n embeds into a unitarily evolving quantum system on $\mathcal{H}$ of dimension $\leq n^3$. P-indivisible transition matrices develop negative entries at fine time resolution; the Stinespring dilation theorem guarantees $\exists$ a Hilbert space and unitary $U(t)$ with $T_{ij}(t) = |U_{ij}(t)|^2$ — transition probabilities are exactly Born-rule probabilities. The proof requires a finite-dimensional configuration space (Axiom 2); Calvo [12] extends the correspondence to infinite dimensions, but this is not required here. An independent derivation using only Stinespring's dilation theorem [32] and standard partial-trace properties [33] is given in Appendix A; either route alone suffices.

From phase space to configuration space. The transition probabilities T_{ij} are projections of the full phase-space flow onto the configuration sector of Γ_V, with momenta and hidden-sector degrees of freedom absorbed into the Liouville marginalization. The resulting process lives on the discrete configuration space of Axiom 2.

Three features emerge. The Schrödinger equation arises from the differentiability of $U(t)$. The Born rule is the equilibrium distribution of the indivisible process [10, 11]. The action scale $\hbar$ enters when defining $\hat{H}(t) \equiv i\hbar\,\partial_t U \cdot U^\dagger$, converting dimensionless rates to energy units; its value requires physical input from the partition (§5).

Phase uniqueness from continuous-time data. The relation $T_{ij}(t) = |U_{ij}(t)|^2$ discards phase information at any single time. Doukas [26] identifies this gap for discrete-time data. The resolution: continuous-time data $\{T_{ij}(t)\}_{t\in\mathbb{R}}$ provides strictly more information.

Lemma (phase-locking). *Let H be Hermitian on $\mathbb{C}^n$ with eigenbasis $\{|k\rangle\}$, eigenvalues $\{E_k\}$, and $V_{ik} = \langle i|k\rangle$. Assume: (G1) non-degenerate spectrum; (G2) non-degenerate energy gaps; (G3) $V_{ik} \neq 0$ for all i, k. Then $T_{ij}(t) = |\langle i|e^{-iHt}|j\rangle|^2$ for all i, j, t uniquely determines H up to an overall energy shift and basis phase conventions.*

Proof. Write:

$$T_{ij}(t) = \sum_{k,l} V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*\; e^{-i(E_k-E_l)t}$$

By (G2), the frequencies $\omega_{kl} = E_k - E_l$ are distinct for distinct pairs with $k \neq l$. Fourier-transforming yields:

$$a_{ij}^{kl} = V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*$$

The moduli $|V_{ik}|$ follow from $a_{ii}^{kl} = |V_{ik}|^2|V_{il}|^2$ (non-zero by (G3)). The phases carry a double-difference structure:

$$\arg(a_{ij}^{kl}) = (\varphi_{ik} - \varphi_{il}) - (\varphi_{jk} - \varphi_{jl})$$

The gauge freedom preserving all double differences is $\varphi_{ik} \to \varphi_{ik} + \alpha_i + \beta_k$ (physically irrelevant basis rephasing). Fixing $\varphi_{1k} = 0$, $\varphi_{i1} = 0$, each remaining phase is $\varphi_{ik} = \arg(a_{i1}^{k1})$,

directly extracted from Fourier data. $H = V\,\mathrm{diag}(E_k)\,V^\dagger$ is unique up to energy shift and phase conventions. □

Conditions (G1)–(G3) fail only on a measure-zero set of Hamiltonians.

1.3.2 3.2 Bell Inequality Violations

The framework is a hidden variable theory evading Bell's theorem [13] not by superdeterminism but by violating factorizability through P-indivisible joint dynamics — while remaining causally local. Two subsystems interacting at preparation carry a joint transition matrix $T_{QR} \neq T_Q \otimes T_R$. This non-factorizability *is* stochastic entanglement [10, 11, 14]. Since the process is indivisible, no well-defined intermediate conditional probabilities permit Bell's factorization.

In the Jarrett decomposition, the framework violates outcome independence while preserving parameter independence and measurement independence — precisely the class Fine's theorem [24] permits. Barandes, Hasan, and Kagan [14] prove the maximum CHSH correlator is exactly Tsirelson's bound $2\sqrt{2}$, with independent support from Le et al. [15]. The bound also follows from the Stinespring route: Appendix A establishes full unitary QM on a tensor-product Hilbert space, from which Tsirelson's original operator-norm argument [36] yields $|\langle S\rangle| \leq 2\sqrt{2}$; achievability follows because all quantum states, including maximally entangled ones, are realizable in the emergent description.

Fine-tuning and causal structure. The framework does not claim to restore Bell locality: outcome independence is genuinely violated. What it provides is a *derivation* of this nonlocality from the causal partition, rather than treating it as axiomatic. The fine-tuning objection (Wood and Spekkens, 2015) assumes DAG causal structure with Markov factorization; P-indivisible processes violate the Markov condition on any DAG, so the appropriate framework is the process tensor [8], within which no-signaling follows from marginalization structure without fine-tuning.

1.3.3 3.3 The Characterization Theorem: Necessity of the Conditions

The logical structure: Barandes' correspondence gives QM $\iff$ P-indivisibility. What remains is P-indivisibility $\implies$ embedded observation under (C1)–(C3).

Dilation existence. Any stochastic process on a finite configuration space can be realized as marginalization of a deterministic process on a larger state space — furnishing Axioms 1–4.

Theorem (C1 equivalence). *The marginalized process is P-indivisible iff T is not a permutation matrix.*

Proof. Forward: §2.3. Reverse: if T is a permutation, $\Lambda(k_2, k_1) = T^{(k_2-k_1)}$ is a valid stochastic matrix for all $k_2 > k_1 \Rightarrow$ P-divisible. □

Theorem (C2 necessity). *In the fast-bath regime ($\tau_B \ll \tau_S$), ergodic mixing drives the hidden sector to uniformity before each coupling event, yielding a Markov chain ($T^{(k)} = T^k$), which is P-divisible. Contrapositively, P-indivisibility requires $\tau_S \ll \tau_B$.*

Proof. Between coupling events (separated by τ_S), the hidden sector evolves under its own Hamiltonian H_H. Let $\mathcal{L}_H$ denote the hidden-sector Liouvillian with spectral gap $\Delta > 0$ (guaranteed on a finite ergodic system). For any conditioned hidden-sector distribution $\mu_H(\cdot|x_i)$ at a coupling event:

$$\|e^{\mathcal{L}_H \tau_S}\mu_H(\cdot|x_i) - \mu_{\mathrm{eq}}\|_{\mathrm{TV}} \leq C\, e^{-\Delta\tau_S}$$

In the fast-bath regime ($\Delta\tau_S \gg 1$), this is exponentially small: the hidden sector relaxes to equilibrium μ_{eq} before each coupling event regardless of its post-coupling state. Each single-

step transition matrix T is therefore computed against the same equilibrium distribution, so $T^{(k)} = T^k$ — a homogeneous Markov chain, hence P-divisible.

Contrapositively: P-indivisibility requires $\Delta\tau_S \lesssim 1$, i.e., $\tau_S \lesssim \tau_B$. For the strong, persistent P-indivisibility of the characterization theorem, the separation must be $\tau_S \ll \tau_B$. (The proof uses discrete coupling events; for continuous Hamiltonian flow, τ_S is the visible sector's dynamical timescale and Δ is the spectral gap of $\mathcal{L}_H$. The continuous-time extension replaces the collision-model structure with the continuous spectral condition $\Delta\tau_S \gg 1 \Rightarrow$ Markovian, established in [4, 9].) □

Theorem (C3 necessity). *Let $m = |\mathcal{C}_H|$. The non-Markovian mutual information satisfies:*

$$I(X_{<t}; X_{>t} \mid X_t) \leq \log_2 m$$

Proof. The total system is deterministic: $X_{>t}$ is a function of (X_t, H_t). Conditioning on X_t: $X_{<t} \to H_t \to X_{>t}$ is a Markov chain. By data processing:

$$I(X_{<t}; X_{>t} \mid X_t) \leq I(X_{<t}; H_t \mid X_t) \leq H(H_t \mid X_t) \leq \log_2 m \quad \square$$

Corollary. P-indivisibility across n visible configurations requires $m \geq n$, and persistence over K transitions requires m/n growing with the observation window — precisely (C3). □

Characterization theorem. *For $|\mathcal{C}_V| \geq 2$, the following are equivalent:*

(i) $\mathcal{S}$ is mathematically equivalent to unitarily evolving QM [10, 11].

(ii) $\mathcal{S}$ is P-indivisible.

(iii) $\mathcal{S}$ arises from marginalizing a deterministic bijection on $\mathcal{C}_V \times \mathcal{C}_H$ with (C1) non-trivial coupling, (C2) slow-bath memory, and (C3) sufficient capacity.

Proof. (i) $\iff$ (ii): Barandes [10, 11]. (iii) $\implies$ (ii): §2.3. (ii) $\implies$ (iii): necessity theorems above. □

QM is not merely *compatible with* embedded observation — it is *equivalent to* it.

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2 PART II: THE COSMOLOGICAL APPLICATION

2.1 4. THE COSMOLOGICAL HORIZON AS CAUSAL PARTITION

2.1.1 4.1 The Partition

The cosmological horizon — the boundary beyond which no signal propagating at or below c can reach the observer — implements the causal partition of Axiom 3: Γ_V is the interior, Γ_H everything beyond. This is a consequence of GR's causal structure, not a modeling choice. The universality of the observed quantum mechanics follows from all sub-c observers sharing the same causal structure.

Different observers have different horizon areas (hence different S_{dS}), but $\hbar = c^3\epsilon^2/(4G)$ depends only on local geometric quantities — not on the observer's worldline or horizon area — so all observers share the same emergent action scale.

2.1.2 4.2 Verification of the Conditions

(C1) In the ADM formulation, the bulk Hamiltonian is a sum of constraints that vanish on-shell; physical time evolution is generated by the boundary term at the horizon, which depends on data from both sides. The Hamiltonian and momentum constraints further correlate interior and exterior initial data on any Cauchy surface, so the physical phase space is a constraint surface within the kinematic product $\Gamma_V \times \Gamma_H$ (see §10.7 for implications). This is *stronger* than the $H_{\text{int}} \neq 0$ required by Axiom 3: constraints enforce correlations that persist on all timescales and cannot be perturbatively removed. **Satisfied.**

(C2) $\tau_B \gtrsim 1/H \sim 10^{17}$ s; for laboratory processes, $\tau_S \sim 10^{-15}$ s. Ratio: $\tau_S/\tau_B \sim 10^{-32}$. **Satisfied.**

(C3) Hidden sector has A/ϵ^2 modes with $A \sim 10^{52}$ m^2; visible sector has $\sim 10^{80}$ baryons. No laboratory process saturates the hidden sector. **Satisfied.**

2.1.3 4.3 Application

Axiom 2 is now a consequence of partition geometry: the horizon has finite area $A = 4\pi c^2/H^2$, bounding coupled modes to $A/\epsilon^2 < \infty$. The Part I theorem applies: the observer's reduced description is P-indivisible and, by Barandes' correspondence, equivalent to unitary QM with $\hbar$ determined by the partition boundary.

2.2 5. DERIVATION OF THE EMERGENT ACTION SCALE ($\hbar$)

2.2.1 5.1 The Classical Horizon Temperature

Jacobson [16] showed $dE = T\,dS$ applied to local causal horizons yields Einstein's equations, with $dE = (c^2\kappa/8\pi G)\,dA$, where κ is the surface gravity of the horizon. The entropy density is $\eta = 1/\epsilon^2$ (one DOF per minimal cell), so $dS = dA/\epsilon^2$. Equating:

$$T_{\text{cl}} = \frac{c^2\epsilon^2\kappa}{8\pi G k_B}$$

For de Sitter ($\kappa = cH$): $k_B T_{\text{cl}} = c^3\epsilon^2 H/(8\pi G)$. No $\hbar$ appears.

2.2.2 5.2 The Emergent Action Scale

Step 1: Uniqueness. By partition-relativity (§1.4), $\hbar$ is derived, not free.

Step 2: $\hbar$ is structural, not volumetric.

Lemma (Boundary-only dependence). *Decompose the hidden sector as $\mathcal{C}_H = \mathcal{C}_B \times \mathcal{C}_D$, where $\mathcal{C}_B$ denotes boundary modes (within the interaction range of H_{int}) and $\mathcal{C}_D$ denotes deep modes (the remainder of the hidden sector). If the classical Hamiltonian is spatially local, then on timescales $t \ll \tau_B$:*

$$T_{ij}(t) = T_{ij}^{(B)}(t) + \mathcal{O}(t/\tau_B)$$

where $T_{ij}^{(B)}$ depends only on the V-B dynamics.

Proof. (i) *Coupling structure.* Spatial locality of the classical Hamiltonian implies the interaction decomposes as $H_{\text{tot}} = H_V + H_B + H_D + H_{VB} + H_{BD}$, with no direct V-D coupling: a deep

hidden-sector mode must propagate through the boundary layer before influencing the visible sector. The coupling chain is $V \leftrightarrow B \leftrightarrow D$, not $V \leftrightarrow D$.

(ii) *Frozen deep sector.* Let Δ_D be the spectral gap of $H_D + H_{BD}$ restricted to the deep sector conditioned on a fixed boundary configuration. The deep-sector evolution over time t displaces any initial configuration by $\mathcal{O}(\Delta_D\, t)$ in phase space. By condition (C2), $\Delta_D \sim 1/\tau_B$, so for $t \ll \tau_B$: $\|\varphi_t^D - \mathrm{id}\| = \mathcal{O}(t/\tau_B)$.

(iii) *Factorization of the Liouville integral.* The transition probability is:

$$T_{ij}(t) = \frac{1}{|\mathcal{C}_B||\mathcal{C}_D|} \sum_{b\in\mathcal{C}_B} \sum_{d\in\mathcal{C}_D} \delta_{x_j}[\pi_V(\varphi_t(x_i, b, d))]$$

By (i), $\pi_V(\varphi_t(x_i, b, d))$ depends on d only through the back-reaction $B \leftarrow D$, which by (ii) shifts b by $\mathcal{O}(t/\tau_B)$. Expanding:

$$\pi_V(\varphi_t(x_i, b, d)) = \pi_V(\varphi_t^{VB}(x_i, b)) + \mathcal{O}(t/\tau_B)$$

where φ_t^{VB} is the flow restricted to $V \times B$ with D frozen. The d-sum then contributes $|\mathcal{C}_D|$ identical terms at leading order:

$$T_{ij}(t) = \underbrace{\frac{1}{|\mathcal{C}_B|} \sum_{b\in\mathcal{C}_B} \delta_{x_j}[\pi_V(\varphi_t^{VB}(x_i, b))]}_{T_{ij}^{(B)}(t)} + \mathcal{O}(t/\tau_B) \qquad \square$$

The correction is $\mathcal{O}(t/\tau_B) \sim 10^{-32}$ for laboratory processes. Since $T_{ij}(t)$ determines the emergent quantum description (§3.1), and $T_{ij}^{(B)}(t)$ depends only on the V-B dynamics — which are characterized by the boundary geometry (A, ϵ, κ) and the constants c, G appearing in the classical Hamiltonian — the emergent action scale $\hbar$ inherits boundary-only dependence.

Step 3: Dimensional determination. With volumetric quantities excluded, the partition-intrinsic quantities are c, G, ϵ. The unique combination with dimensions of action:

$$\hbar = \beta \frac{c^3\epsilon^2}{G}$$

Step 4: Thermal self-consistency. The classical substratum assigns the partition boundary a temperature $k_B T_{\mathrm{cl}} = c^3\epsilon^2\kappa/(8\pi G)$ (§5.1), containing no $\hbar$. The emergent QFT of Part I lives on this classical background, which has a bifurcate Killing horizon with surface gravity κ. Regularity of the Wick-rotated metric at the horizon requires Euclidean period $\beta = 2\pi c/\kappa$; any QFT on this background — including a lattice-regularized one — must therefore be periodic in imaginary time with the same period, giving a KMS state at temperature $T_Q = \hbar\kappa/(2\pi c k_B)$, with $\hbar$ unknown. This is a theorem *within* the derived QFT, not an external import. A thermometer at the boundary is a visible-sector object whose click rate is computable in either description; the classical and emergent accounts of one boundary cannot disagree, so $T_{\mathrm{cl}} = T_Q$:

$$\frac{c^3\epsilon^2\kappa}{8\pi G} = \frac{\hbar\kappa}{2\pi c}$$

κ cancels (confirming the structural/volumetric distinction). Solving:

$$\boxed{\hbar = \frac{c^3\epsilon^2}{4G}}$$

The derivation is a gap equation: ϵ is the free geometric input, $\hbar$ is the output. The two temperatures are computed independently: $T_{\rm cl}$ from the classical substratum alone (Axioms 1–4, the Jacobson identity, and $\eta = 1/\epsilon^2$), T_Q from the emergent QFT alone (Part I's theorem plus Euclidean regularity), with $\hbar$ as the sole unknown. The match is non-trivial: $T_{\rm cl}$ could depend on volumetric hidden-sector data (excluded by Step 2), or T_Q could depend on the state (it does not — the KMS temperature is purely kinematic). That neither pathology obtains makes the gap equation a genuine determination.

Jacobson's original derivation [16] uses $\hbar$-containing forms; this paper does not — it uses the classical identity $dE = (c^2\kappa/8\pi G)\, dA$ and the classical entropy density $\eta = 1/\epsilon^2$. The Gibbons-Hawking temperature [17] $k_B T_{\rm GH} = \hbar\kappa/(2\pi c)$ is recovered as a prediction, not used as an input. The KMS condition used above is derived for continuum QFT on bifurcate Killing horizons; for the lattice-regularized theory emerging from Part I, the relevant result is that thermal periodicity of the Euclidean Green's functions is robust against UV modifications of the dispersion relation, with corrections at $\mathcal{O}((\epsilon\kappa/c)^2)$ [28, 29]. For the cosmological horizon, $\epsilon\kappa/c = \epsilon H/c \sim 10^{-61}$, so lattice corrections are $\sim 10^{-122}$.

2.2.3 5.3 Gauge-Fixing and the Dimensional Obstruction

Steps 1–2 establish that $\hbar$ depends only on boundary quantities; Step 4 will fix its value via thermal matching. A prior question must be addressed: does the transition-probability data $\{T_{ij}(t)\}$ determine the emergent Hamiltonian uniquely, or does residual gauge freedom leave $\hbar$ underdetermined? The following theorem, completing the phase-locking argument of §3.1, shows the gauge freedom is physically trivial.

Theorem (D-gauge completeness). *Let $U(t) = e^{-iHt}$ with non-degenerate eigenvalues and non-vanishing configuration-basis overlaps. If $|U'_{ij}(t)|^2 = |U_{ij}(t)|^2$ for all i, j, t, then $H' = DHD^\dagger$ for a time-independent diagonal unitary D.*

Proof. Write $U_{ij}(t) = \sum_a V_{ia}\, e^{-iE_a t}\, V^*_{ja}$ with $V_{ia} = \langle i|a\rangle$ and eigenvalues $\{E_a\}$, and similarly $U'_{ij}(t) = \sum_a V'_{ia}\, e^{-iE'_a t}\, V'^*_{ja}$.

Step 1 (Eigenvalue recovery). By the phase-locking argument of §3.1, $|U_{ij}(t)|^2 = |U'_{ij}(t)|^2$ for all t implies, via Fourier analysis, that the frequency sets $\{E_a - E_b\}$ and $\{E'_a - E'_b\}$ coincide. Non-degeneracy of energy gaps then gives $E'_a = E_{\sigma(a)} + E_0$ for some permutation σ and global shift E_0. The Fourier coefficients $a^{kl}_{ij} = V_{ik}V^*_{jk}V_{jl}V^*_{il}$ must match between primed and unprimed, which (by the non-vanishing condition on overlaps) forces $\sigma = \mathrm{id}$ up to relabeling. Thus $E'_a = E_a + E_0$.

Step 2 (Modulus recovery). From the diagonal Fourier coefficients $a^{kl}_{ii} = |V_{ik}|^2|V_{il}|^2$, the non-vanishing condition gives $|V'_{ia}| = |V_{ia}|$ for all i, a.

Step 3 (Phase structure). Write $V'_{ia} = V_{ia}\, e^{i\delta_{ia}}$. The off-diagonal Fourier coefficients require:

$$V'_{ik}V'^*_{jk}V'_{jl}V'^*_{il} = V_{ik}V^*_{jk}V_{jl}V^*_{il}$$

Substituting and canceling moduli:

$$e^{i(\delta_{ik}-\delta_{jk}-\delta_{il}+\delta_{jl})} = 1 \quad \text{for all } i, j, k, l$$

This is the double-difference condition: $\delta_{ik} - \delta_{il} - \delta_{jk} + \delta_{jl} = 0 \pmod{2\pi}$. The general solution on the integer lattice is $\delta_{ia} = \alpha_i + \beta_a$ for phases $\{\alpha_i\}$, $\{\beta_a\}$. Thus $V' = D_\alpha V D_\beta$ where $D_\alpha = \mathrm{diag}(e^{i\alpha_i})$ and $D_\beta = \mathrm{diag}(e^{i\beta_a})$. The D_β factor is absorbed into the eigenvector phase convention (physically irrelevant), giving $H' = D_\alpha H D_\alpha^\dagger$ with $D = D_\alpha$ a time-independent diagonal unitary. □

The D-gauge is physically trivial (basis rephasing). Cross-interval transition probabilities fix the relative gauge up to a single global phase; in the continuum limit this becomes an unobservable smooth function $e^{i\theta(t)}$.

Dimensional obstruction. The unitary $U(t)$ is dimensionless; $\hbar$ enters only via $\hat{H} = i\hbar\,\partial_t U \cdot U^\dagger$. No dimensionless data can fix a dimensionful constant — Step 4's thermal self-consistency provides the necessary dimensionful input.

2.3 6. SELF-CONSISTENCY AND THE DISCRETENESS SCALE

Rearranging $\hbar = c^3\epsilon^2/(4G)$:

$$\epsilon^2 = \frac{4\hbar G}{c^3} = 4\,l_p^2$$

The framework yields $\epsilon = 2\,l_p$ exactly. This is algebraically equivalent to the formula for $\hbar$: the framework contains one free geometric parameter (ϵ), and thermal self-consistency establishes its relationship to $\hbar$. The result is a self-consistency condition constraining ϵ to the Planck regime, not an independent determination. If $\epsilon^2 \ll l_p^2$, sub-Planckian subcells would create a second trace-out, breaking the single-valuedness of $\hbar$. If $\epsilon^2 \gg l_p^2$, the emergent description would assign distinct quantum states to physically identical configurations. The framework's predictive content comes from the relationship $\hbar = c^3\epsilon^2/(4G)$, the Bekenstein-Hawking formula, and the consequences that follow.

With $\epsilon = 2\,l_p$, the number of independent modes on the cosmological horizon is:

$$N_{\text{modes}} = \frac{A}{\epsilon^2} = \frac{A}{4\,l_p^2} = S_{\text{dS}}$$

where $S_{\text{dS}} = A/(4\,l_p^2)$ is the Bekenstein-Hawking entropy of the de Sitter horizon. The factor of 1/4 in the Bekenstein-Hawking formula — historically introduced as a proportionality constant — is here derived: each minimal cell of area $\epsilon^2 = 4\,l_p^2$ on the horizon contributes one unit of entropy, and $A/(4\,l_p^2) = A/\epsilon^2$ recovers the standard formula exactly.

2.4 7. DISSOLUTION OF THE COSMOLOGICAL CONSTANT PROBLEM

2.4.1 7.1 The Two Levels of Description

From finite-dimensional QM to QFT. Part I delivers QM on a finite-dimensional Hilbert space. The $N = A/\epsilon^2$ cells discretize the visible sector; spatial locality of the classical substratum restricts which cells interact on any given timescale. The emergent Hilbert space decomposes as $\mathcal{H} = \bigotimes_k \mathcal{H}_k$ — a lattice-regularized QFT with UV cutoff at $\epsilon = 2\,l_p$. The tensor product structure follows from the spatial product structure of the classical configuration space $\mathcal{C}_V =$

$\mathcal{C}_1 \times \cdots \times \mathcal{C}_N$ (one factor per cell), which is not required by Axiom 2 but is a consequence of spatial locality in the classical substratum: each cell's configuration is a local degree of freedom.

Locality preservation theorem. *If the classical Hamiltonian is spatially local (couples only neighbors), then the emergent quantum Hamiltonian inherits spatial locality.*

Proof. For infinitesimal dt and $x' \neq x$: $T_{x,x'}(dt) = |\langle x'|\hat{H}|x\rangle|^2\, dt^2 + O(dt^3)$. If x, x' differ at non-neighboring sites, spatial locality of the classical dynamics gives $T_{x,x'}(dt) = 0$ at leading order, hence $|\langle x'|\hat{H}|x\rangle|^2 = 0$. By the D-gauge theorem (§5.3), the residual phase freedom is a diagonal unitary D that preserves the vanishing of off-diagonal moduli, so $\langle x'|\hat{H}|x\rangle = 0$ in any gauge. The emergent Hamiltonian has the form $\hat{H} = \sum_k \hat{h}_k + \sum_{\langle k,l\rangle} \hat{h}_{kl}$. □

Scope of the emergent QFT. The CC dissolution requires only that the emergent description assigns zero-point energy $\sim \hbar\omega/2$ per mode with UV cutoff at ϵ^{-1} — a property shared by any lattice QFT with local couplings, regardless of gauge group or matter content. The framework's predictions in Part II ($\hbar$, S_{dS}, ν_{OI}) depend on partition geometry, not on the emergent field content.

Classical substratum (what geometric measurements probe): The horizon has classical thermal equilibrium. By the Friedmann equation, $\rho_{\mathrm{crit}} = 3H^2c^2/(8\pi G)$. No zero-point energy, no discrepancy. Total vacuum energy density: $\rho \sim H^2/G \sim 10^{-9}$ J/m^3 — precisely observed.

Emergent QFT (what local quantum measurements probe): Zero-point energy $\frac{1}{2}\hbar\omega$ per mode summed to the Planck cutoff gives $\rho_{\mathrm{QFT}} \sim 10^{113}$ J/m^3 — a $\sim 10^{122}$ discrepancy with the observed value [2, 3, 18].

2.4.2 7.2 Why Gravity Sees the Classical Value

Spacetime geometry is part of the classical substratum (Axiom 3): the metric evolves via Einstein's equations *before* the trace-out. The stress-energy sourcing gravity is the classical stress-energy of the total microstate, not the expectation value of an emergent quantum operator. The semiclassical equation $G_{\mu\nu} = 8\pi G\langle\hat{T}_{\mu\nu}\rangle$ is itself an emergent approximation; at the classical level, the gravitational field equations never encounter the zero-point sum.

Ontological commitment. Classical spacetime is fundamental; QM is emergent. This ordering is not a modeling choice but is forced by the logical structure of the derivation through three independent requirements:

(i) Definiteness. The trace-out that produces QM is defined by a *specific* partition $\Gamma = \Gamma_V \times \Gamma_H$. The partition must be definite — not in superposition, not subject to quantum uncertainty — for the marginalization integral (§1.4) to be well-defined. A partition in superposition would yield an incoherent mixture of inequivalent quantum theories, not a single QM. The metric at the partition boundary must therefore be classical.

(ii) Non-circularity. The partition is defined by the causal structure: Γ_H is the set of degrees of freedom beyond the observer's causal reach (§4.1). Causal structure is determined by null geodesics of the metric. If the metric were derived from QM, then the derivation would require: QM → metric → causal structure → partition → QM — a circular dependency with no entry point. The circle breaks only if the metric exists prior to the partition.

(iii) Uniqueness of $\hbar$. The emergent action scale is determined by the boundary geometry (§5.2). If the geometry were itself quantum-mechanical, $\hbar$ would depend on a quantum state — but $\hbar$ is state-independent by observation. The geometry-first ordering is the only one consistent with a universal $\hbar$.

Any framework in which QM is logically prior to the metric must either (a) treat the zero-point sum as a gravitational source — producing the 10^{122} discrepancy requiring cancellation or fine-

tuning — or (b) invoke an independent mechanism to decouple quantum vacuum energy from gravity. In the present framework, the zero-point sum is an artifact of the emergent description and never enters the stress-energy tensor; no decoupling mechanism is needed.

Cumulative evidence for the ordering. The geometry-first ordering produces six independent consequences that match observation, none of which were designed into the axioms: (1) the CC dissolution — the observed vacuum energy sits at $\rho \sim H^2/G$, the classical baseline, without fine-tuning (§7.3); (2) the Bekenstein-Hawking formula including the 1/4 factor, derived from $\epsilon = 2\,l_p$ (§6); (3) the value of $\hbar$, determined by partition geometry rather than treated as a free parameter (§5.2); (4) the dark-sector concordance — ~95% of ρ_{crit} is invisible to QFT, matching the observed dark sector (§9); (5) dark energy evolution with ν_{OI} in the range consistent with DESI data (§8.1); (6) a testable GW echo prediction at the discreteness scale (§8.2). Quantum-first orderings produce the 10^{122} discrepancy, have no natural account of the dark sector, and treat $\hbar$ and the 1/4 factor as unexplained inputs. The cumulative case is not a single tiebreaker but a pattern: every quantitative output of the geometry-first ordering is consistent with observation; the central quantitative output of the quantum-first ordering is the worst prediction in physics.

The classical vacuum energy scale. The framework does not explain *why* $\rho \sim H^2/G$; this is set by initial conditions via the Friedmann equation. What it does is reclassify the CC puzzle: from quantum vacuum cancellation (10^{122}-fold, unsolved for five decades) to classical initial conditions (a standard cosmological question).

Quantum corrections to gravity. The emergent quantum description does feed back into gravitational dynamics through state-level quantities: §8.1 derives dark energy evolution because the emergent vacuum energy depends on H through the hidden sector's volume. What the framework excludes is the zero-point sum — a property of the correspondence itself rather than of any particular state — as a gravitational source. The structural/volumetric distinction of §5.2 applies: $\hbar$ (structural) does not gravitate; the vacuum energy (volumetric, state-dependent) does, but at the classical scale $\rho \sim H^2/G$ rather than $\rho \sim M_{\text{Pl}}^4$.

2.4.3 7.3 The Structural Origin of the 10^{122} Discrepancy

$\rho_{\text{QFT}}/\rho_{\text{classical}} \sim M_{\text{Pl}}^4/(H^2/G) = S_{\text{dS}} = A/(4\,l_p^2)$ — the Bekenstein-Hawking entropy, now identified as the information compression ratio of the trace-out. The "worst prediction in physics" is the entropy of the observer's blind spot — a category error, not a fine-tuning failure. Wolpert's limits of inference [19] confirm this cannot be resolved from within: both geometric and quantum measurements are faithful to their respective descriptions, and no embedded observer can determine which is more fundamental. This is not a prediction awaiting future data — the observed vacuum energy sits at the classical geometric scale $\rho \sim H^2/G$, the value the framework expects.

2.5 8. PREDICTIONS AND TESTS

2.5.1 8.1 Dark Energy Evolution in Running Vacuum Form

$\hbar$ is H-independent (§5.2), but the emergent vacuum energy is a state-level quantity depending on the hidden sector's volume. Expanding $\Lambda_{\text{eff}}(H)$:

$$\Lambda_{\text{eff}}(H) = \Lambda_0 + \frac{3\nu_{\text{OI}}}{8\pi G}(H^2 - H_0^2) + \mathcal{O}((H^2 - H_0^2)^2)$$

This is the Running Vacuum Model [20, 21]. The $S_{\rm dS}$ horizon modes span frequencies $\omega_{\rm IR} = H$ to $\omega_{\rm UV} = c/\epsilon$. For conformal fields, the spectral density is uniform in $\ln\omega$, so the number of independent spectral channels is $\mathcal{N}(H) = \ln(c/\epsilon H)$. Each carries fraction $1/\mathcal{N}$ of the total gravitating vacuum energy, so $\rho_\Lambda \propto 1/\mathcal{N}$. Taylor-expanding:

$$\boxed{\nu_{\rm OI} = \frac{\Omega_\Lambda}{2\,\ln(c/\epsilon H_0)}}$$

With $\epsilon = 2\,l_p$, $H_0 = 67.4$ km/s/Mpc, $\Omega_\Lambda = 0.685$: $\nu_{\rm OI} = 2.45 \times 10^{-3}$ (parametric uncertainty $\pm 0.03 \times 10^{-3}$ from H_0 and Ω_Λ; systematic range $\sim$ 1–5 $\times 10^{-3}$ from the spectral assumption below). Independently testable ratio: $\nu_{\rm OI}/\Omega_\Lambda \approx 0.00358$.

Sensitivity to field content. The conformal assumption (uniform spectral density in $\ln\omega$) is an idealization; massive fields and QCD confinement break conformal invariance. For a spectral density $\propto \omega^\alpha$ with $\alpha \neq 0$, the effective channel count $\mathcal{N}$ is modified. Since $\nu_{\rm OI} \propto 1/\mathcal{N}$ and the conformal value is $\mathcal{N} \sim 140$, even order-unity changes to α shift $\nu_{\rm OI}$ within $\mathcal{O}(10^{-3})$: the logarithmic structure suppresses sensitivity to UV field content. The systematic range 1–5$\times 10^{-3}$ spans a factor-of-two variation in $\mathcal{N}$ around the conformal value; this is the dominant uncertainty, not the parametric error. The central value 2.45×10^{-3} should be understood as the conformal benchmark, not a precision prediction.

DESI data [22, 30] report evolving dark energy at 2.8σ–4.2σ. RVM fits [31] find $\nu \sim \mathcal{O}(10^{-3})$ with $\nu = 0$ disfavored at 2.7σ–3.1σ, consistent with the prediction.

2.5.2 8.2 Gravitational Wave Echoes

A black hole horizon represents a regime where the causal partition differs from the cosmological one: degrees of freedom reclassify from visible to hidden sector as they cross the horizon, shifting the effective partition and the emergent description.

At proper distance $\sim \epsilon$ outside a black hole horizon r_h, an infalling mode reaches the discreteness floor and must scatter back. The predicted time delay is:

$$\Delta t_{\rm echo} \approx \frac{r_h}{c} \ln\left(\frac{r_h}{2\,l_p}\right)$$

For a $30M_\odot$ remnant, logarithmic dependence on ϵ makes this robust ($\sim$ 0.1 ms shift within the $\sim$ 54 ms timescale [23]). Detection would constitute direct evidence that ϵ has observable consequences at horizons other than the cosmological one — a regime where partition-relativity (§10.2) becomes empirically accessible. Joint confirmation of dark energy evolution *and* GW echoes would uniquely favor an information-theoretic origin, since standard RVMs predict $\Lambda(H)$ running but have no mechanism for discreteness-scale echoes.

2.5.3 8.3 The Dark Sector as Corroboration

The trace-out that produces QM has a gravitational consequence: the boundary entropy is invisible to the emergent QFT but gravitationally active. Part III shows that this accounts for the observed fact that ~95% of the universe's gravitational content has no source in particle physics — providing independent corroboration of the framework's structural claims.

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3 PART III: THE DARK SECTOR AS CORROBORATION

3.1 9. THE INVISIBLE GRAVITATIONAL BUDGET

Observationally, ~95% of the universe's gravitational content — the dark sector comprising dark energy (~68%) and dark matter (~27%) — has no source in quantum field theory. The baryonic sector accounts for only ~5% of ρ_{crit}. This section shows that the trace-out producing QM simultaneously and unavoidably renders the boundary entropy gravitationally active but invisible to the emergent description, with total energy density equal to ρ_{crit}. The framework does not set out to explain the dark sector; it produces the dark sector as an automatic consequence of the same mechanism that produces quantum mechanics.

Part II identified the 10^{122} cosmological constant discrepancy as the *information compression ratio* of the trace-out — the cost of describing the visible sector in the emergent quantum language. This section identifies a complementary quantity: the *gravitational occlusion fraction* — the share of the universe's gravitational budget that the trace-out renders invisible to the emergent description. The compression ratio measures how much the emergent description distorts what the observer can see; the occlusion fraction measures how much of what gravitates the observer cannot see at all. Together, they account for the two largest anomalies in modern cosmology as two aspects of a single phenomenon: the cost of observing the universe from within.

3.1.1 9.1 The Entropy Medium

Lemma (Entropy medium density). *The gravitational potential generated by the de Sitter boundary entropy at the Gibbons-Hawking temperature equals, at any interior point, the potential of a uniform medium at the critical density.*

Proof. The horizon carries $S_{\text{dS}} = A/(4\,l_p^2) = A/\epsilon^2$ modes (§6). The thermal self-consistency of §5.2 establishes $T_{\text{GH}} = T_{\text{cl}}$; critically, the classical temperature $T_{\text{cl}} = c^3\epsilon^2 H/(8\pi G k_B)$ is computed entirely within the classical substratum (§5.1) — the Jacobson identity, the entropy density $\eta = 1/\epsilon^2$, and the surface gravity $\kappa = cH$ — with no reference to $\hbar$ or the emergent quantum description. The total thermal energy is $E_s = S_{\text{dS}} \cdot k_B T_{\text{cl}}$. Distributing this over the enclosed volume $V_H = 4\pi c^3/(3H^3)$ gives the effective density:

$$\rho_s = \frac{E_s}{V_H} = \frac{A}{\epsilon^2} \cdot \frac{c^3\epsilon^2 H}{8\pi G} \cdot \frac{3H^3}{4\pi c^3} = \frac{3c^2H^2}{8\pi G} = \rho_{\text{crit}}$$

By the shell theorem (Gauss's law for gravity), a uniform mass shell of total energy E produces, at any interior point, the same gravitational potential as a uniform medium of density E/V filling the enclosed volume. The de Sitter isometry ensures the boundary entropy is uniformly distributed on the horizon, so the shell theorem applies. □

Four clarifications are essential for what follows.

Classical, not emergent. The calculation above uses only pre-trace-out quantities: the number of boundary modes (A/ϵ^2), the classical temperature (T_{cl}), and the Hubble volume (V_H). No step invokes the emergent quantum description. This is what distinguishes ρ_s from the QFT zero-point energy $\rho_{\text{QFT}} \sim M_{\text{Pl}}^4$: the latter is computed as $\sum \frac{1}{2}\hbar\omega$ over modes of the *emergent* quantum field theory — a quantity that exists only after the trace-out. The framework denies that ρ_{QFT} gravitates (§7.2), because it is an artifact of the emergent description. The boundary entropy's thermal energy, by contrast, is a classical quantity that predates the trace-out and gravitates at the classical level. That both involve $\hbar$ numerically (because $T_{\text{cl}} = T_{\text{GH}}$) is a consequence of the gap equation's self-consistency, not a dependence on QM. The ratio $\rho_{\text{QFT}}/\rho_s \sim S_{\text{dS}} \sim 10^{122}$

— the same compression ratio identified in §7.3 — confirming that the two quantities occupy different levels of the framework's hierarchy.

Physical locus. The entropy modes are physically localized on the horizon — they are the hidden-sector boundary degrees of freedom of §5.2. The uniform-medium description is a gravitational equivalence (shell theorem), not a claim about physical delocalization. No entropy occupies the bulk.

Why the equivalence is useful. When matter is present, it locally deforms the horizon geometry (§9.2–9.3), redistributing the boundary entropy. The gravitational effect of this redistribution on bulk observers is most easily computed using the uniform-medium equivalence: the deformed boundary is equivalent to a non-uniform effective bulk source. Whether this redistribution produces observable local effects (beyond the uniform background already captured by the Friedmann equation) depends on the redistribution profile — a question addressed in §9.4.

Algebraic status. The coincidence $\rho_s = \rho_{\text{crit}}$ is an algebraic consequence of the Friedmann equation and the self-consistency relation $\hbar = c^3\epsilon^2/(4G)$; it is not an independent physical input.

This identity, which Verlinde [37] assumes, is here derived.

3.1.2 9.2 The Matter-Boundary Interaction

Theorem (Invisibility). *The gravitational influence of the boundary entropy has no operator in the emergent QFT.*

Proof. The argument has two logically distinct components.

(i) Construction-level invisibility (from Part I alone). The boundary modes comprising S_{dS} are hidden-sector degrees of freedom — the variables traced out to produce the quantum description. By the trace-out structure, no hidden-sector degree of freedom maps to an operator on $\mathcal{H}_V$. This is a theorem of Part I, independent of any gravitational content: the hidden sector is invisible to the emergent description by construction.

(ii) Gravitational consequence (from the ontological ordering of §7.2). The physical import — that the boundary entropy's gravitational influence appears as additional gravity with no source in $\langle \hat{T}_{\mu\nu} \rangle$ — requires the geometry-first ordering established in §7.2: the metric exists at the pre-trace-out level, and the stress-energy sourcing gravity is the classical stress-energy, not the emergent quantum expectation value. Under this ordering, the gravitational interaction between bulk matter and boundary entropy is mediated by the metric at the classical level, modifying the geometry but contributing no term to the emergent $\langle \hat{T}_{\mu\nu} \rangle$. To the embedded observer, this appears as additional gravity with no particle-sector source.

Part III's predictions are therefore conditional on the same geometry-first ordering that dissolves the CC problem in §7.2. This is not an additional assumption: the ordering is already required by Part II's logic (the partition is defined by null geodesics, so the metric must exist prior to the partition). What Part III provides is a second, qualitatively distinct *consequence* of that ordering — not a second independent derivation of it. □

In the regime where baryonic gravity dominates ($g_b \gg cH$), the boundary entropy's influence is a small perturbation. The partition boundary's surface gravity cH is the only acceleration scale intrinsic to the partition geometry, suggesting a transition at $g_b \sim cH$; but whether the redistribution produces significant local effects — and if so, at what precise scale — requires a quantitative force law beyond the axioms (§9.4).

3.1.3 9.3 Memory Effects and P-Indivisibility

Matter locally deforms the horizon geometry, redistributing the boundary entropy (§9.1). This displacement persists because $\tau_B \gtrsim 1/H$ (condition C2): the hidden-sector modes carrying the redistributed entropy do not thermalize back to the unperturbed configuration on sub-Hubble timescales. Verlinde [37] identifies this as a "memory effect" from the non-thermalization of de Sitter states at sub-Hubble scales. In the present framework, this non-thermalization is not assumed but derived: it is P-indivisibility operating in the gravitational sector, with (C2) ensuring that entropy displacements persist without thermal overwriting.

3.1.4 9.4 The Concordance

Theorem (Invisible gravitational budget). *Under Axioms 1–4 and conditions (C1)–(C3), the cosmological trace-out that produces quantum mechanics simultaneously produces a gravitational budget with the following properties: (i) total effective energy density ρ_{crit} (§9.1); (ii) no source in the emergent QFT (§9.2); (iii) persistent entropy displacements by matter, producing history-dependent gravitational structure (§9.3).*

Proof. Each property is proved in the referenced subsection. □

The framework accounts for the baryonic sector (~5% of ρ_{crit}) through the emergent QFT. The remaining ~95% is the boundary entropy — gravitationally active but invisible to the emergent description. Crucially, this ~95% gravitates because the boundary entropy's thermal energy is a classical quantity computed before the trace-out (§9.1, first clarification), unlike the QFT zero-point energy which is an artifact of the emergent description and does not gravitate (§7.2). This matches the observed composition of the universe, in which ~95% of the gravitational content (dark energy + dark matter) has no source in particle physics. The uniform component of the boundary entropy corresponds to dark energy (addressed in §7); the matter-induced redistribution (property (iii)) is a candidate mechanism for the structured component (dark matter), though the redistribution profile — and hence the dark energy/dark matter split — requires a quantitative force law beyond Axioms 1–4 [37], deferred to future work.

The concordance is non-trivial. The axioms were formulated to address embedded observation, not cosmological composition. That they produce a gravitational budget matching the observed dark sector is independent corroboration that observational incompleteness is capturing real physical structure. The parallel with the CC dissolution is exact: the 10^{122} compression ratio (§7.3) and the ~95% occlusion fraction are both mandatory consequences of the same trace-out, addressing the two largest anomalies in cosmology through a single mechanism.

Falsifiability. The invisible-budget prediction would be falsified by detection of dark matter particles accounting for the full ~27% (contradicting (ii)), or by discovery that the dark sector's gravitational effects are unrelated to the boundary geometry.

3.2 10. DISCUSSION

3.2.1 10.1 Interpretive Consequences

In the cosmological application, the hidden sector is the trans-horizon region; the degrees of freedom involved in quantum experiments — photons, electrons, slits, detectors — are all visible-sector objects. Their quantum behavior is a downstream consequence of the single cosmological trace-out: the emergent quantum mechanics, once established by the theorem, governs all visible-sector dynamics.

In the double-slit experiment, the particle traverses a single slit in the deterministic substratum. The interference pattern arises because opening or closing the second slit changes the boundary conditions of the transition matrix, altering the distribution of detection events. A which-path detector at one slit couples the trajectory to additional visible-sector degrees of freedom, changing the transition matrix and eliminating the interference terms. In Wigner's friend, the Friend has a definite outcome; Wigner's superposition reflects his epistemic deficit. The Everettian measure problem dissolves: the system evolves as a single reality; "branches" are features of the compressed description. The Born rule, often treated as an independent postulate, is derived: it is the equilibrium distribution of the indivisible stochastic process (§3.1), not an additional assumption.

3.2.2 10.2 Scope

The characterization theorem is a full equivalence: QM $\iff$ embedded observation under (C1)–(C3). The theorem does not identify which physical systems satisfy the conditions (empirical) nor derive $\hbar$ for specific systems (requires Part II). Universality in our universe follows from the shared causal partition defined by null geodesics.

3.2.3 10.3 The Stochastic-Quantum Bridge

The bridge is established by two independent routes: Barandes' stochastic-quantum correspondence [10, 11] (peer-reviewed, corroborated [12, 25, 26]) and the Stinespring construction (Appendix A, using only textbook results [32, 33]). Either alone suffices; together they overdetermine the bridge. Partition-relativity, emergent stochasticity, and P-indivisibility (§§1.4, 2) invoke neither.

3.2.4 10.4 The Status of the Discreteness Scale

ϵ does not smuggle in a quantum assumption. Axiom 2 requires finite-dimensionality, motivated by the classical fact that finite-area boundaries admit finitely many modes. The result $\epsilon = 2\,l_p$ is a *consequence* of self-consistency (§6). Holographic entropy bounds [27] provide independent, non-quantum motivation.

3.2.5 10.5 Vacuum Energy: Relative Effects and the Higgs Potential

Relative effects. Casimir forces and Lamb shifts are predictions *within* the emergent QFT — relative effects depending on energy differences between configurations. The framework denies that the *absolute* zero-point sum gravitates, because gravity operates at the logically prior classical level (§7.2).

The Higgs potential. The strongest CC objection is not the zero-point sum but the electroweak Higgs potential: symmetry breaking shifts $V(\phi)$ by $\sim$ (200 GeV)^4, exceeding the observed Λ by $\sim$ 55 orders of magnitude. The framework's response follows from its ontological ordering: the Higgs field, like all quantum fields, is part of the emergent description (§7.1). The classical substratum has deterministic dynamics on a finite configuration space; the degrees of freedom that the emergent theory organizes as a scalar field with spontaneous symmetry breaking have a non-field-theoretic character in the substratum. The vacuum energy shift is therefore a property of the emergent ground state — the same category as the zero-point sum. The dissolution applies uniformly: if QM is emergent, then so is the Higgs mechanism and its vacuum energy.

3.2.6 10.6 Relation to Prior Work

't Hooft [1]: Differs in mechanism (epistemic trace-out vs. information loss), generality (any system satisfying axioms vs. particular rules), and Bell placement (outcome independence violation vs. superdeterminism). **QBism/relational QM:** Share observer-relative epistemic states; this framework provides a structural *why* and quantitative predictions. **Jacobson [16]:** Derives gravity from thermodynamics; this framework derives QM from causal structure — complementary, with mutual consistency as a check.

Danielson-Satishchandran-Wald (DSW) horizon decoherence [38, 39]. DSW proved within standard QFT that any Killing horizon imparts a fundamental rate of decoherence on nearby quantum superpositions: the horizon registers the long-range field of the superposed body via soft gravitons/photons, destroying coherence at a rate determined by the horizon geometry. This is precisely the framework's partition mechanism described in QFT language — the horizon is the causal boundary, information leaks through it, and the exterior state decoheres. Three specific features match: (i) the decoherence rate depends on horizon geometry (partition-relativity); (ii) decoherence vanishes for extremal black holes where the coupling to the interior is screened (C1 failure); (iii) the process is non-Markovian near the horizon, with information backflow from the interior [40]. The DSW program derives these results within the emergent QFT; the present framework explains *why* horizons produce decoherence — because the trace-out across a causal partition is the mechanism that generates quantum mechanics itself.

AdS/CFT and the geometry-entanglement connection. The Ryu-Takayanagi formula, Van Raamsdonk's disconnection argument, ER=EPR, and "it from qubit" programs all suggest entanglement is prior to geometry. The framework offers a uniform alternative reading: tracing out across a geometric boundary produces entanglement entropy proportional to boundary area (A/ϵ^2 modes), so each of these results is equally consistent with geometry assembling entanglement rather than the reverse. For Ryu-Takayanagi specifically, the proportionality constant must follow from the partition structure — and it does: §6 derives the 1/4 factor. In the holographic context, the boundary CFT observer is *external* to the bulk, immune to the inferential limits of embedded observers [19]; the framework does not claim to reproduce holographic results in that setting.

The point of contact is empirical. Entanglement-first programs predict spacetime breakdown at low entanglement or high energy. This framework predicts QM breakdown near $\epsilon = 2\,l_p$ while the geometric substratum persists. The geometry-first ordering produces six independent concordances with observation (§7.2); entanglement-first orderings inherit the zero-point sum as a gravitational source and have no natural account of the dark sector.

Witten's Type II algebra program achieves finite entropy without finite-dimensional Hilbert spaces; this is compatible, since the framework's predictions depend on finite entropy (the physical content of Axiom 2), not on dimensional finiteness per se.

3.2.7 10.7 Assumptions, Limitations, and Falsifiability

The theorem requires Axiom 2 (independent in Part I; consequence of geometry in Part II), the stochastic-quantum bridge (§10.3; established by two independent routes), and genericity conditions (non-degenerate spectrum, non-vanishing overlaps), which hold for all but a measure-zero set. The P-indivisibility proof relies essentially on finiteness (the recurrence argument uses $\varphi^N = \mathrm{id}$); the theorem is therefore a statement about finite systems. Its applicability to our universe depends on Part II's identification of a physical ϵ from the horizon's finite area.

The sharp partition and constraint structure. Axiom 3's product decomposition $\Gamma = \Gamma_V \times \Gamma_H$ is the kinematic phase space. In GR, the Hamiltonian and momentum constraints restrict the

physical phase space to a submanifold $\Gamma_{\text{phys}} \subset \Gamma_V \times \Gamma_H$ on which interior and exterior data are correlated. P-indivisibility survives on a constraint surface because restoring P-divisibility would require decoupling the sectors entirely, violating the constraints; the constraint surface thus enforces (C1) non-perturbatively. For systems other than the cosmological horizon, the fidelity of the product approximation determines the fidelity of the emergent quantum description.

Planck-scale ordering. The classical-prior ontology may break down at l_p, where "geometric" and "quantum" lose operational distinction. If a deeper theory unifies both, this framework describes the effective macroscopic regime.

Falsifiability. The *theorem* would be falsified by a failure of both the stochastic-quantum correspondence and the Stinespring construction for the class of processes generated here — a possibility excluded by the independent proofs in §3.1 and Appendix A. The *cosmological application* would be falsified by definitive absence of dark energy evolution or GW echoes at the predicted timescale. The *dark-sector concordance* (§9.4) would be falsified by detection of dark matter particles accounting for the full structured dark sector, or by evidence that the dark sector's gravitational effects are unrelated to the cosmological boundary geometry.

Open problems. The framework does not address the regime of nested or overlapping partitions (e.g., a black hole horizon inside the cosmological horizon), where two partition boundaries with different geometries coexist. §8.2 treats the specific case of GW echoes; the general theory requires further development. Laboratory tests of the characterization theorem are possible in analogue gravity systems where the hidden sector capacity is tunable; detailed predictions for BEC analogue black holes will be presented separately.

3.3 11. CONCLUSION

The general theorem (Part I). Under four axioms and conditions (C1)–(C3), the P-indivisibility of reduced dynamics is proved: any bijective dynamics on a finite system with non-trivial coupling necessarily produces P-indivisible stochastic dynamics. By the stochastic-quantum correspondence, the observer's description is necessarily quantum mechanics. Continuous-time transition data resolves all gauge freedom. The Schrödinger equation, Born rule, and Bell violations are structural consequences. The characterization theorem establishes that the conditions are necessary: QM and embedded observation under (C1)–(C3) are equivalent.

The cosmological application (Part II). The cosmological horizon provides a realization where the sharp-partition approximation is exact and all conditions hold. The theorem yields: (a) $\hbar = c^3\epsilon^2/(4G)$ with $\epsilon = 2\,l_p$ and the Bekenstein-Hawking formula as consequences; (b) dissolution of the CC problem — the 10^{122} discrepancy is S_{dS}, the compression ratio of the observer's quantum description; and (c) falsifiable predictions including RVM dark energy with $\nu_{\text{OI}} \approx 2.45 \times 10^{-3}$ and gravitational wave echoes.

The dark sector as corroboration (Part III). The trace-out that produces quantum mechanics simultaneously renders ~95% of the universe's gravitational budget invisible to the emergent QFT: the boundary entropy has effective energy density ρ_{crit}, no QFT operator, and matter-induced displacements that persist through P-indivisibility. This gravitational occlusion fraction parallels the 10^{122} information compression ratio of Part II: one measures how much the trace-out distorts the emergent vacuum energy, the other how much of the gravitational universe the trace-out hides. Together they account for the cosmological constant problem and the dark sector — the two largest anomalies in modern cosmology — as two aspects of a single phenomenon.

The apparent conflict between quantum mechanics and general relativity, the 10^{122} vacuum energy discrepancy, and the dark sector are three faces of the same fact: the information-theoretic cost of observing the universe from within.

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3.6 APPENDIX A: INDEPENDENT DERIVATION VIA STINESPRING DILATION

This appendix derives the emergent quantum description using only Stinespring's dilation theorem [32] and standard properties of the partial trace [33], without invoking the stochastic-quantum correspondence [10, 11]. The two routes are logically independent; either alone suffices.

3.6.1 A.1 Setup

The finite configuration spaces $\mathcal{C}_V = \{x_1, \ldots, x_n\}$ and $\mathcal{C}_H = \{h_1, \ldots, h_m\}$ (Axiom 2) embed into Hilbert spaces $\mathcal{H}_V = \mathbb{C}^n$ and $\mathcal{H}_H = \mathbb{C}^m$ via $|i\rangle \leftrightarrow x_i$ and $|k\rangle \leftrightarrow h_k$. This introduces no quantum postulates: it is the canonical identification of probability distributions on a finite set with diagonal density matrices.

3.6.2 A.2 Permutation Unitarity

Lemma A.1. *Any bijection $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ defines a unitary U_φ on $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_H$.*

Proof. Define $U_\varphi|i,k\rangle = |\varphi(x_i, h_k)\rangle$. Since φ is a bijection, U_φ permutes the orthonormal basis, hence is unitary. □

For continuous-time dynamics φ_t, Stone's theorem on $\mathcal{H}$ yields $U_t = e^{-i\hat{H}t}$ for Hermitian $\hat{H}$.

3.6.3 A.3 The Quantum Channel

The observer's ignorance of the hidden sector (Axioms 3–4) corresponds to $\rho_H = I_m/m$. The visible-sector quantum channel is

$$\Phi(\rho_V) = \mathrm{Tr}_H \left[U_\varphi \left(\rho_V \otimes \rho_H \right) U_\varphi^\dagger \right]$$

This is CPTP by a standard result [33, Theorem 8.1], with Kraus representation $\Phi(\rho_V) = \sum_{k,l} K_{kl}\, \rho_V\, K_{kl}^\dagger$ where $K_{kl} = m^{-1/2}\langle l|U_\varphi|k\rangle_H$. The triple $(\mathcal{H}_H, U_\varphi, \rho_H)$ is the Stinespring dilation [32] of Φ.

3.6.4 A.4 Born Rule Recovery

Theorem A.2. *The classical transition probabilities T_{ij} (§1.4) equal the Born-rule probabilities of Φ.*

Proof. $P(j|i) = \langle j|\Phi(|i\rangle\langle i|)|j\rangle = m^{-1}\sum_{k,l} |\langle j,l|U_\varphi|i,k\rangle|^2$. Since U_φ is a permutation, $\langle j,l|U_\varphi|i,k\rangle = \delta_{(j,l),\varphi(i,k)}$. Thus $P(j|i) = m^{-1}\sum_k \delta_{j,\pi_V(\varphi(x_i,h_k))} = T_{ij}$. □

3.6.5 A.5 Emergent Coherence

Theorem A.3. *If T is not a permutation matrix (condition (C1)), then Φ generates genuine quantum coherence: it is not entanglement-breaking.*

Proof. If T is not a permutation, some initial state $|i\rangle$ maps to at least two distinct outputs, so $\Phi(|i\rangle\langle i|)$ has rank ≥ 2. By linearity, Φ maps some off-diagonal input $|i\rangle\langle j|$ to a non-zero operator, precluding the measure-and-prepare form of entanglement-breaking channels. □

3.6.6 A.6 CP-Indivisibility

Theorem A.4. *The P-indivisibility of §2.3 implies CP-indivisibility of $\{\Phi_t\}$: there exist $t_2 > t_1 > 0$ with no CPTP map Λ satisfying $\Phi_{t_2} = \Lambda \circ \Phi_{t_1}$.*

Proof. CP-divisibility restricted to diagonal inputs reduces to P-divisibility. The contrapositive gives the result. □

By the Breuer-Laine-Piilo criterion [4, 6], CP-indivisibility implies non-monotonic trace distance (information backflow) — the quantum signature of non-Markovianity.

3.6.7 A.7 Approximate Unitarity

On observable timescales $t \ll \tau_B$, the hidden-sector state is approximately frozen (conditions (C2)–(C3)). The channel generator decomposes as $d\Phi_t/dt|_{t=0}(\rho_V) = -i[\hat{H}_{\text{eff}}, \rho_V] + \mathcal{D}(\rho_V)$ with $\mathcal{D} \sim \mathcal{O}(\tau_S/\tau_B)$. The leading-order dynamics is the Schrödinger equation. The phase-locking lemma (§3.1) and D-gauge theorem (§5.3) then determine $\hat{H}_{\text{eff}}$ uniquely from continuous-time transition data.

3.6.8 A.8 Comparison of Routes

	Barandes route (§3.1)	Stinespring route (this appendix)
Input	P-indivisible process on $\mathcal{C}_V$	Bijection on $\mathcal{C}_V \times \mathcal{C}_H$ + marginalization
Bridge	Stochastic-quantum correspondence [10, 11]	Stinespring dilation [32] + partial trace [33]
Output	Unitary with $T_{ij} = \|U_{ij}\|^2$	CPTP channel with $T_{ij} = \langle j\|\Phi(\|i\rangle\langle i\|)\|j\rangle$
Born rule	Equilibrium of indivisible process	Partial trace structure (Theorem A.2)
Scope	Any P-indivisible process	Processes from marginalized bijections

The Barandes route is more general; the Stinespring route requires only textbook results [32, 33] predating 2000. Together they establish that the bridge is overdetermined.